



3 Complex Analysis 2024

Tutorial 1

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Complex numbers: basics

Problem 1 Find the real and the imaginary parts of the complex numbers

$$z_1 = (3 + 2i)^3 \quad \text{and} \quad z_2 = \frac{i + 1}{i - 1}.$$

Problem 2 Prove the following inequalities for $z, w \in \mathbb{C}$:

- (a) $\max\{|\operatorname{Re} z|, |\operatorname{Im} z|\} \leq |z| \leq |\operatorname{Re} z| + |\operatorname{Im} z|,$
- (b) $||w| - |z|| \leq |w - z|.$

Problem 3 Find all $z \in \mathbb{C}$ with the following property:

- (a) $z = |z|,$
- (b) $|z - i| = 2z + i.$

Problem 4 Sketch the following sets in the complex plane:

- (a) $\{z \in \mathbb{C} : \operatorname{Re}(z - 1) \geq 0\},$
- (b) $\{z \in \mathbb{C} : \operatorname{Im}((1 + i)z) < 0\},$
- (c) $\{z \in \mathbb{C} : 2 < |z| < 4\},$
- (d) $\{z \in \mathbb{C} : |z - i| \leq 2 \text{ and } \operatorname{Im} z \geq 2\}.$

Problem 5 Consider the equation

$$z^5 + 32 = 0.$$

- (a) Plot the solutions of this equation in the complex plane.
- (b) Give the solutions in polar coordinates.